Jangho Lee

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PROFESSIONAL PROFILES

- Google Scholar: https://scholar.google.com/citations?user=wBEE2YAAAAAJ
- ORCID: https://orcid.org/0000-0002-8942-1092
- LinkedIn: https://www.linkedin.com/in/jholee92/
- GitHub: https://github.com/jangholee92

RESEARCH INTERESTS

GeoAI; Climate AI; Generative AI; Urban Climate; Land Modelling; Climate Downscaling; Environmental Data Science; Nature Based Solution; Urban Natural Hazard; Remote Sensing; Climate Change; Climate Informatics

PROFESSIONAL EXPERIENCE

New York University | ***Courant Institute of Mathematics, Computing, and Data Science*** ***2025 - Present***
Postdoctoral Researcher New York, NY, USA

- Developed AI-based downscaling algorithm as part of the ACCLIMITISE project
- Spearheaded a data fusion project integrating remote-sensing, in-situ and climate model outputs
- Led multi-nation research projects, including countries in Africa and Europe

University of Illinois Chicago | ***Department of Earth and Environmental Sciences*** ***2023 - 2025***
Postdoctoral Researcher Chicago, IL, USA

- CROCUS Artificial Intelligence and Digital Twins (AIDT) initiative planning lead
- Led multiple projects regarding urban natural hazard, nature-based solution, and GeoAI
- Directed multi-institute collaboration to publish 10+ peer reviewed papers

EDUCATION

Texas A&M University | ***Atmospheric Sciences*** ***2018 - 2023***
Doctor of Philosophy *in Atmospheric Science* | Advisor: Dr. Andrew E. Dessler College Station, TX, USA

- Thesis: *Extreme Temperature Events Caused by Climate Change and Variability: Drivers and its Impact*

Seoul National University | ***School of Earth and Environmental Sciences*** ***2011 - 2018***
Bachelor of Science *in Earth and Environmental Science* | Advisor: Dr. Kwang-Yul Kim Seoul, South Korea

- Thesis: *Analysis of Source Regions and Meteorological Factors for the Variability of Spring PM10 Concentrations in Seoul, Korea* [Best Thesis Award]

PUBLICATIONS

Citations: 348 | H-Index: 12 *(as of 10/15/2026)*

Published Peer-Reviewed Papers: 28 | Lead Author (First and/or Corresponding): 24 | co-Author: 4

***** *Corresponding Author* | $^{+}$ *Equal Contribution* | $^{\xi}$ *Mentored Students*

2026

1. **Lee, J*** & Shamekh, S (2026) Evaluation of Conditional Diffusion for Cross-sensor Land Surface Temperature Downscaling Across African Cities. *International Journal of Remote Sensing*, 47 (20)
2. **Lee, J*** & Shamekh, S (2026) Model Architecture and Calibration Period Sensitivity in Controlled Super-Resolution Reconstruction of Near-Surface Air Temperature. *Machine Learning: Earth*, 2 (2), 025017

3. Muradyan, P*., Collis, S., O'Brien, J. R., Tuftedal, M., Jackson, R., Raut, B. A., Grover, M., Kaludi, B., Vincent, A. E. S., Wagner, T. J., Sankaran, R., Cho, A., Weart, B., Pal, S., Sullivan, R. C., Anderson, G., Berkelhammer, M., Conrad, N., DuMerer, D., Fytanidis, D. K., Gala, T., Gonzalez-Meler, M. A., Hence, D., **Lee, J**., Lozano, F., Michnick, J., Miller, W. M., Nesbitt, S., Packman, A. I., Park, S. Y., Saari, C., Shahkarami, S., Wang, J., Wawrzyniak, E., Yocca, D., Zvolanek, E., Beaudry, S., Kotamarthi, R., Negri, C (2026) CROCUS Micronet: A Distributed, AI-Enabled Urban Observation System in Chicago. *Bulletin of the American Meteorological Society* [In Press]
4. **Lee, J*** & Berkelhammer, M (2026) Spatiotemporal Resoinse of Urban Bike-Sharing Ridership to Weather, Air Quality, and Future Climate change in Major US Cities. *Journal of Environmental Management*, 414, 130431
5. **Lee, J*** & Shamekh, S (2026) How Far Can We Downscale? Resolution Limits and Physical Interpretability of Diffusion Models for African Precipitation. *Machine Learning: Earth*, 2 (1), 015018
6. **Lee, J***, Berkelhammer, M., O'Brien, J., McNicol, G., Vincent, A.E.S., Grover, M., Packman A.I., Kaludi, B., Cho, A., Gonzalez-Meler, M (2026) Imputation of Urban Environmental Sensor Data using Gated Attention Bidirectional Long Short-Term Memory (GA-BiLSTM): Methods, Performance, and Implications. *Environmental Monitoring and Assessment*, 198, 262
7. **Lee, J***, Berkelhammer, M., Vincent, A.E.S., Grover, M., Cho, A., Packman, A.I., Kaludi, B., Gonzalez-Meler, M., McNicol, G (2026) Quality Assessment and Control of Urban Environmental Sensors using Physical Thresholding and Machine Learning-based Probabilities. *Big Earth Data*, 1-27

2025

8. **Lee, J*** (2025) Online Signals of Vulnerability: Revealing City-Level Response to Extreme Heat Using Google Trends. *Environmental Research Letters*, 20, 124034.
9. **Lee. J***., Park, S. Y+., Wadhwa, A+., Packman, A., Nesbitt, S. W., Sharma, A., Berkelhammer, M., Garcia, M. H., Kotamarthi, R., Hence, D., Miller, W. M (2025) Comparing Multi-Source Urban Flood Indicators: Satellite, Simulation, and Citizen-Reported Data. *Environmental Research: Water*, 1(3), 035007
10. Kotamarthi, R*., Wang, J., Stock, J., Fytanidis, D. K., Munson, T., Niyogi, D., Muradyan, P., Jackson, R., Collis, S., Gonzalez-Meler, M. A., **Lee, J**., Sharma, A., Wang, S., Kaludi, B., Kaplan, M., Nesbitt, S. W., Martilli, A., Jacob, R., Negri, C (2025) Artificial Intelligence-Enabled Digital Twin for U.S. Cities. *Bulletin of the American Meteorological Society*. 106 (11), E2411-E2418
11. **Lee. J***. & Berkelhammer, M (2025) Evaluating the Influence of Traffic Congestion on Surface Urban Heat Island Intensity. *Geophysical Research Letters*, 52(16), e2025GL116651.
12. **Lee, J***., Berkelhammer, M., Park, S. Y., Wilson, M (2025) Analysis of Urban Flooding in Chicago Based on Crowdsourced Data: Drivers and the Need for Community-Based Mitigation Strategies. *Environmental Research: Infrastructure and Sustainability*, 5 (2), 025008
13. Cho, A*., Love, N., Cintron, R., Nicholson, J., Xu, L., Nunez-Mir, G., **Lee, J**., Berkelhammer, M., Gonzalez-Meler, M (2025) Plant Species Selection and Participatory Community Co-design are Essential in Balancing Ecosystem Services and Disservices in Urban Areas. *Environmental Research Letters*, 20, 051003.
14. **Lee, J*.** & Park, S.Y. (2025) WGAN-GP-Based Conditional GAN (cGAN) with Extreme Critic for Precipitation Downscaling in a Key Agricultural Region of the Northeast U.S. *IEEE Access–Geoscience and Remote Sensing Society Section*, 13, 46030-46041.
15. **Lee, J*.** (2025) Inferring Urban Air Temperatures from Land Surface Temperatures with the E3SM Land Model (uELM), Satellite Observations, and Measurement Campaign. *IEEE Access–Geoscience and Remote Sensing Society Section*, 13, 32564-32573.
16. **Lee, J*.** (2025) Estimating Near-Surface Air Temperature from Satellite-Derived Land Surface Temperature Using Temporal Deep Learning: A Comparative Analysis. *IEEE Access–Geoscience and Remote Sensing Society Section*, 13, 28935-28945.

2024

17. **Lee, J*.** & Berkelhammer, M. (2024) Observational Constraints on the Spatial Effect of Greenness and Canopy Cover on Urban Heat in Major Midlatitude City. *Geophysical Research Letters*, 51(1), e2024GL110847.
18. Cho, A*., Dziedzic, N., Davis, A., Hanson, C., **Lee, J.**, Nunez-Mir, G., Gonzalez-Meler, M. A. (2024). Leaf Functional Traits Highlight Phenotypic Variation of Two Tree Species in the Urban Environment. *Frontiers in Plant Science Functional Plant Ecology*, 15, 1450723.
19. **Lee, J*.** (2024). Assessment of U.S. Urban Surface Temperature using GOES-16 and 17 Data: Urban Heat Island and Temperature Inequality. *Weather, Climate, and Society,* 16(2), 315-329.

20. **Lee, J**[*]., Berkelhammer, M., Wilson, M. D., Love, N., & Cintron, R. (2024). Urban Land Surface Temperature Downscaling in Chicago: Addressing Ethnic Inequality and Gentrification. *Remote Sensing*, 16(9), 1639.
21. **Lee. J**[*]. & Hu, M. (2024). Effect of Environmental and Socioeconomic Factors on Increased Early Childhood Blood Lead Levels: A Case Study in Chicago. *International Journal of Environmental Research and Public Health*, 21, 383.
22. **Lee, J.** & Dessler, A. E[*]. (2024). Improved Surface Urban Heat Impact Assessment Using GOES Satellite Data: A Comparative Study With ERA-5. *Geophysical Research Letters*, 51(1), e2023GL107364.

Prior to 2024

23. **Lee, J**[*]**.** & Dessler, A. E. (2023). Future Temperature-Related Deaths in the US: The Impact of Climate Change, Demographics, and Adaptation. *GeoHealth*, *7*(8), e2023GH000799.
24. **Lee, J**[*]**.** & Dessler, A. E. (2022). The Impact of Neglecting Climate Change and Variability on ERCOT's Forecasts of Electricity Demand in Texas. *Weather, Climate, and Society*, *14*(2), 499-505.
25. **Lee, J.**, Mast, J. C., & Dessler, A. E[*]. (2021). The Effect of Forced Change and Unforced Variability in Heat Waves, Temperature Extremes, and Associated Population Risk in a CO_2-Warmed World. *Atmospheric Chemistry and Physics*, *21*(15), 11889-11904.
26. **Lee, J.**, Shi, Y. R., Cai, C., Ciren, P., Wang, J., Gangopadhyay, A., & Zhang, Z[*]. (2021). Machine Learning Based Algorithms for Global Dust Aerosol Detection from Satellite Images: Inter-Comparisons and Evaluation. *Remote Sensing*, *13*(3), 456.
27. **Lee, J.** & Kim, K. Y[*]. (2018). Analysis of Source Regions and Meteorological Factors for the Variability of Spring PM10 Concentrations in Seoul, Korea. *Atmospheric Environment*, *175*, 199-209.
28. **Lee, J**[*]**.** (2017). Future Trend in Seasonal Lengths and Extreme Temperature Distributions over South Korea. *Asia-Pacific Journal of Atmospheric Sciences*, *53*, 31-41.

FUNDED GRANTS & AWARDS

- Google Cloud Platform Research Grant [Co-PI, 20,000 USD]: *Diffusion-Based Multi-Sensor Satellite Fusion for Land Surface Temperature* (2026)
- 1st Place in Climate Investment Challenge [Co-PI, 5,000 GBP]: *AquaRisk Horizon* (2026)

PATENTS

- Method of Prediction: Mercury Concentration in Blood (South Korea Patent #1019623150000)

TALKS & PRESENTATIONS (Recent 3 years)

- A Multiscale Model-Driven Experimental (ModEx) Framework for Urban Areas – a CROCUS approach, *AGU*, 2025
- The 2025 CROCUS Urban Flooding and Rainfall Campaign: *International Conference on Urban Climate*, 2025
- Urban Land Surface Temperature Downscaling in Chicago: Addressing Ethnic Inequality and Gentrification, *AGU*, 2024
- Comparing Multi-source Urban Flood Indicators: Satellite, Simulation, and Citizen-reported Data, *DOE UIFL Meeting*, 2024
- Observational Constraints on the Spatial Effect of Greenness and Canopy Cover on Urban Heat in a Major Midlatitude City, *DOE UIFL Annual Meeting*, 2024
- Urban Land Surface Temperatures: Importance, Measurements, and Applications, *Florida State University*, 2024 [Invited]
- Urban Land Surface Temperatures: Measurements, and Applications, *University of Illinois Chicago*, 2024 [Invited]
- Urban Land Surface Temperature Downscaling in Chicago: Addressing Disparities, *Seoul National University*, 2024 [Invited]

RECOGNITIONS & MISCELLANEOUS

- Selected in AGU Global Environmental Change Early Career Researcher Spotlight, 2025
- Top Cited Paper in GeoHealth, AGU, 2025
- Outstanding Graduate Student Research Award, Texas A&M University, 2021
- Outstanding Graduate Student Seminar Award, Texas A&M University, 2021
- Best Thesis Award, Seoul National University, 2017
- Merit-Based Scholarship, Seoul National University, 2011-2017

PYTHON PACKAGES, BOOKS & EDUCATION MATERIALS

- STELAR (Spatio-TEmporaL lAg-based Regression) GAM: https://github.com/jangholee92/stelar_gam
- Python and Statistics for Climate and Atmospheric Informatics (Book in Prep): https://github.com/jangholee92/pythonForCI
- E3SM & ELM Documentation with OLMT: https://github.com/jangholee92/ELM_Tutorial

TECHNICAL PROFICIENCY

- Proficient in Python, R, and Linux
- Proficient in ML, DL, and HPC Modules: TensorFlow, Keras, Scikit-Learn, XGBoost, Dask
- Proficient in AI algorithms and architectures: GAN, Diffusion Models, Autoencoders, etc.
- Proficient in GIS tools: QGIS, ArcGIS, GeoPandas, Google Earth Engine (GEE)
- Proficient in land-atmospheric modelling (ELM)

RESIDENTIAL STATUS

- Lawful Permanent Resident (LPR) of the United States
- South Korea Citizen